MILITARY SPECIFICATION

ALIGNMENT DEVICE, CARTRIDGE TO ATOMIC WARHEAD: H4168

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of alignment device entitled Alignment Device, Cartridge to Atomic Warhead: H4168 which is used with the Nike Hercules warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.

DRAWINGS

ORDNANCE CORPS

8837379	- Case, Carrying For H4168 and H4169.
8837382	- Box Packing, Ammunition for H4168 Alignment Devices Cartridge to Warhead and H4169 Wrench, Spanner to Carrying Case.
8838738	- Alignment Device, Cartridge to Atomic Warhead: H4168.
8857293	- Pin, Shoulder, Headed.
8857294	- Bushing, Sleeve.
8857295	- Washer, Flat.
8857296	- Adapter, Alignment Device, Cartridge to Atomic Warhead H4168.
8857297	- Frame, Alignment Device.
8857299	- Pin, Straight, Headless.
8857300	- Key, Machine.
8857301	- Collar, Shaft.
EL-8856709	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The alignment device shall comply with all requirements specified on the drawing (dwg) 8838378, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot formation.

4.2.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting

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safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.2.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.2.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.2.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.2.2.1.1 Pin, straight, headless (see dwg 8857299 covering a detail of dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Diameter, minimum (min.)	Gage
Minor:		
	201. Length incorrect	Gage
	202. Evidence of poor workmanship (see 3.3)	Visual

4.2.2.1.2 Pin, shoulder, headed (see dwg 8857293 covering a detail of dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Distance from taper to lower undercut, incorrect	Gage
	102. Distance from lower undercut to shoulder, incorrect	Gage
	103. Diameter below upper undercut, max	Gage
	104. Diameter above upper undercut, max	Gage
	105. Diameter above taper, max	Gage
	106. Total length incorrect	Gage
Minor:		
	201. Finish inadequate	Visual
	202. Chamfer missing or incorrect	Visual
	203. Evidence of poor workmanship (see 3.3)	Visual

4.2.2.1.3 Bushing, sleeve (see dwg 8887294 covering a detail of dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Small outside diameter, min.....Gage
102. Inside diameter, max.....Gage
103. Length to shoulder, min.....Gage

Minor:
201. Total length, max.....Gage
202. Evidence of poor workmanship (see 3.3).....Visual

4.2.2.1.4 Adapter, alignment device, cartridge to atomic warhead: H4168 (see dwg 8857296 covering a detail of dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread, min.....	Gage
102.	Minor diameter of thread, max.....	Gage
103.	Depth of threaded cavity, min.....	Gage
104.	Small outside diameter.....	Gage
105.	Thickness of large diameter, min.....	Gage
Minor:		
201.	Large outside diameter, min.....	Gage
202.	Finish inadequate.....	Visual

4.2.2.1.5 Washer, flat (see dwg 8857295 covering a detail of dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Thickness, max.....	Gage
102.	Outside diameter, min.....	Gage
103.	Inside diameter, max.....	Gage
Minor:		
201.	Finish incorrect.....	Visual
202.	Evidence of poor workmanship (see 3.3).....	Visual

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4.2.2.1.6 Collar, shaft (see dwg 8857301 covering a detail of dwg 8838738).

Categories		Characteristics	Method of Inspection
Critical:		None defined.	
Major:			
	101.	Diameter of four 0.2660 dia. holes in collar	Gage
	102.	Diameter of counterbore in four small holes in collar, min	Gage
	103.	Diameter of two 0.2506 dia. holes in collar, min	Gage
	104.	Symmetry of two 0.2506 holes in collar	Gage
	105.	Length of collar, max	Gage
	106.	Diameter of 1.770 hole in shaft, min	Gage
	107.	Diameter of four 0.1248 holes in shaft	Gage
	108.	Width of keyway, max	Gage
Minor:			
	201.	Width of collar	Gage
	202.	Chamfer missing	Visual
	203.	Undercut missing	Visual
	204.	Finish inadequate	Visual
	205.	Evidence of poor workmanship (see 3.3)	Visual

4.2.2.1.7 Key, machine (see dwg 8857300 covering a detail of dwg 8838738).

Categories		Characteristics	Method of Inspection
Critical:		None defined.	
Major:			
	101.	Width, max	Gage
	102.	Long chamfer incorrect	Gage
	103.	Width of protrusion at side, max	Gage
Minor:			
	201.	Undercut missing	Visual
	202.	Evidence of poor workmanship (see 3.3)	Visual

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4.2.2.1.8 Frame, alignment device, cartridge to atomic warhead: H4168 (see dwg 8857297 covering a detail of dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter of two holes at ends	Gage
102.	Symmetry of two holes at ends	Gage
103.	Diameter of counterbore in two small holes, max	Gage
104.	Depth of counterbore in two small holes, min	Gage
105.	Minor diameter of threads, max	Gage
106.	Pitch diameter of threads, max	Gage
107.	Diameter of large hole, min	Gage
108.	Diameter of counterbore in large hole, min	Gage
109.	Depth of counterbore in large hole, max	Gage
Minor:		
201.	Width at center, incorrect	Gage
202.	Width at ends, incorrect	Gage
203.	Length, incorrect	Gage
204.	Thickness, incorrect	Gage
205.	Evidence of poor workmanship (see 3.3)	Visual

4.2.2.1.9 Alignment device, cartridge to atomic warhead: H4168 after assembly of key (see dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter of holes in key	Gage
Minor:	None defined.	

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4.2.2.1.10 Alignment device, cartridge to atomic warhead: H4168 (see dwg 8838738).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Marking missing, misleading or unidentifiable	Visual
102.	Pins fit improperly	Visual/Manual
103.	Any component part fails to fit properly	Visual/Manual
104.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Engraving missing, incorrect or mislocated	Visual
202.	Evidence of poor workmanship (see 3.3)	Visual

4.2.2.1.11 Case, carrying for XH4168 (align device) and XH4169 (spanner wrench) prior to closing (see dwg 8837379 covering a detail of dwg 8837382).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Component missing or not properly pack	Visual
102.	Pad missing, insufficient or partially detached	Visual
103.	Wire missing or broken	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.3)	Visual

4.2.2.1.12 Case, carrying for XH4168 (align device) and XH4169 (spanner wrench) closed (see dwg 8837379, covering a detail of dwg 8837382).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Latches missing partially detach.....Visual
102. Latches un-operative (cannot secure box).....Visual
103. Handle missing or partially detached.....Visual
104. Box damaged to extent it exposes contents.....Visual
105. Nail protruding.....Visual

Minor:
201. Marking missing, misleading or unidentifiable.....Visual
202. Home plate missing or partially detached.....Visual
203. Hinge butt missing or not properly secure.....Visual
204. Protective coating missing.....Visual

4.2.2.1.13 Box, packing, ammunition for XH4168 alignment device and XH4169 (spanner wrench), opened (see dwg 8837382).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Carrying case missing.....Visual

Minor:
201. Barrier bag torn, perforated.....Visual
202. Fillers missing or insufficient.....Visual
203. Evidence of poor workmanship (see 3.3).....Visual

4.2.2.1.14 Box, packing, ammunition for XH4168 (alignment device) and XH4169 (spanner wrench) closed (see dwg 8837382).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Box damaged to extent it exposes contents.....Visual
102. Connector missing or partially detached.....Visual
103. Strapping missing, broken or loose.....Visual
104. Car seal missing or insecured.....Visual

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Minor:
201. Marking missing, misleading or unidentifiable.....Visual
202. Coating inadequate.....Visual
203. Evidence of poor workmanship (see 3.3).....Visual

4.3.2.1.15 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with (see dwg 8737382).....	Visual
	102. Marking not in accordance with (see dwg 8797272).....	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.3).....	Visual

4.2.3 Testing. There are no tests applicable to the alignment device.

4.2.4 Acceptance inspection equipment.

4.2.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8856709). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.2.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.3 Test methods. There are no test methods applicable to the alignment device.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The alignment device shall be packaged in accordance with dwg 8737382 using the specified moisture barrier.

5.1.2 Level C. The alignment device shall be packaged in accordance with dwg 8737382 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The alignment device shall be packed in accordance with dwg 8737382.

5.3 Marking. Marking shall be as specified in accordance with dwg 8797272.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.2.4.1). Data specified in 4.2.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.2.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description X-PA-PD-2247 dated 19 June 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1190-A092

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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